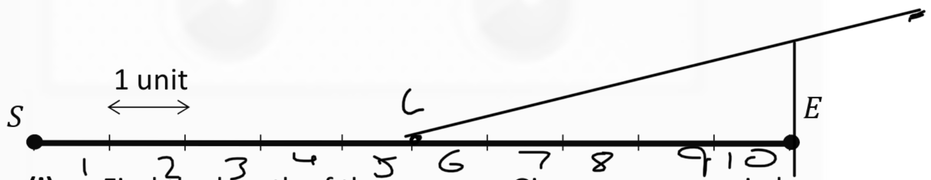
Question 9 **(50 marks)**

The line segment [SE], shown below, represents an airport runway.
The point S and the point E represent the start and end points of the runway respectively.
The diagram is drawn to a scale of 1 unit = 250 metres.



(a) **(i)** Find the length of the runway. Give your answer in km.

$250 \times 10 = 2500$ metres

2.5 km

(ii) An aircraft starts at the point S and travels 1 250 m to a point L where it lifts off.
From the point L, the aircraft makes an angle of $14°$ with the ground, [LE].
The aeroplane travels along this path until it is directly above E.
Plot **and** label the lift-off point L, and **draw** the aircraft's flight path for this part of its journey, on the diagram above.

(iii) Find the total distance the aeroplane has travelled when it is directly above E.
Give your answer, in metres, correct to the nearest metre.

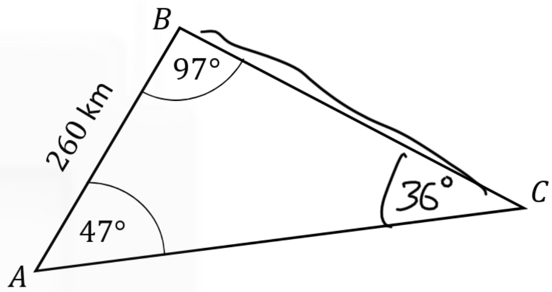
$\cos 14 = \dfrac{1250}{h}$

$h(\cos 14) = 1250$

$h = \dfrac{1250}{\cos 14}$

1288 m

(b) The aircraft flies from airport A to airport B, and then on to airport C, at the same altitude. The pilot records the flight summary on the given diagram.



(i) Find the distance from airport B to airport C. Give your answer correct to the nearest km.

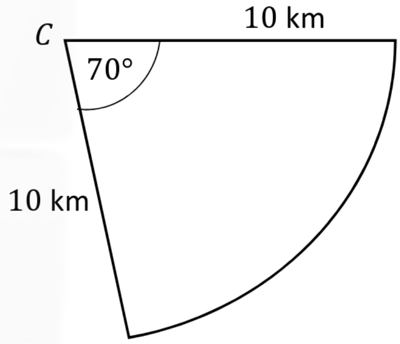
$$\frac{a}{\sin 47} = \frac{260}{\sin 36}$$

$$a = \frac{260(\sin 47)}{\sin 36}$$

$$a = 323.5$$

324 km

(ii) When the plane was directly over airport C, the pilot was instructed to "circle" until a runway was available. She therefore flew 10 km away from C before turning and flying along a circular arc of 70° and then returning to the airport.
This path was all flown at the same altitude.



Find the total distance travelled.
Give your answer, in km, correct to two decimal places.

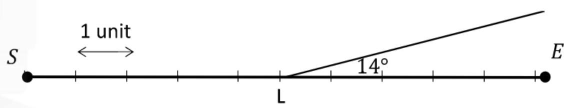
$$\pi r^2 \left(\frac{\theta}{360}\right)$$

$$\pi 10 \left(\frac{70}{360}\right)$$

12.21 km

$12.21 + 10 + 10$

$\Rightarrow 32.21$ km

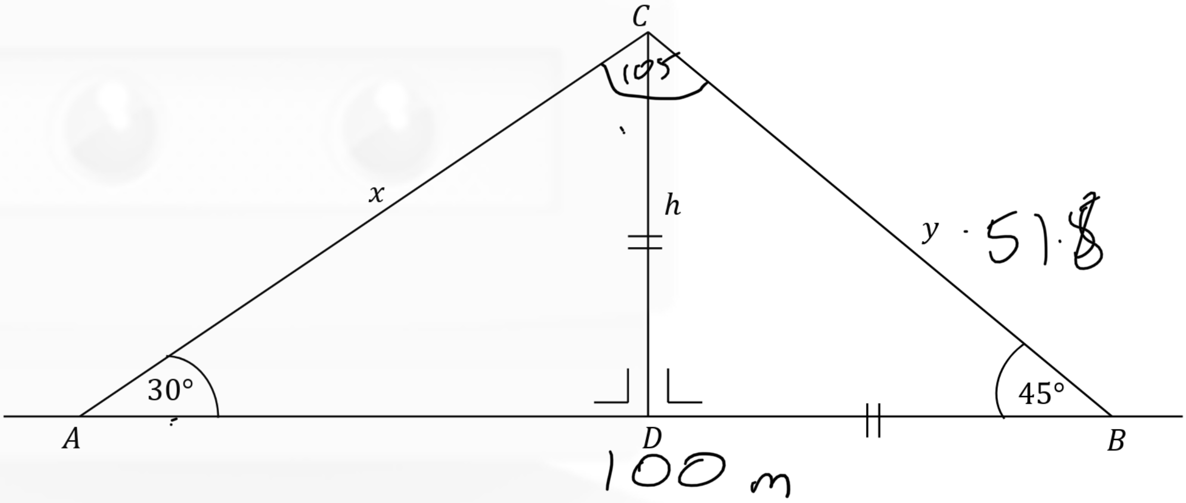
	(a) **(i)**	$250 \times 10 = 2500$ $= \dfrac{2500}{1000}$ $= 2{\cdot}5$	**Scale 10C (0, 3, 7, 10)** *Low Partial Credit:* Work of merit • Numbering units • Incorrect work with 250 or 10 *High Partial Credit:* Significant work • $250 \times 10 = 2500$ • Error in converting to km *Full Credit:* • Correct answer without supporting work
	(a) **(ii)**		**Scale 5B (0, 2, 5)** *Partial Credit:* Work of merit • L marked but in incorrect position • L marked correctly but angle incorrect **Notes**: • Correct work but L not indicated merits (F*) **Note:** Tolerance $\pm 1°$

(a) (iii)	$\cos 14° = \frac{1250}{h}$ $\Rightarrow \text{h} = 1288{\cdot}267$ $\text{h} = 1288$ $\text{Total Distance} = 1288 + 1250$ $= 2538$	**Scale10D (0, 3, 5, 8, 10)** *Low Partial Credit:* Work of merit • Correct trigonometric ratio or relevant formula with some substitution • Indicates addition of 1250 *Mid Partial Credit:* • Formula fully substituted $(\cos 14° = \frac{1250}{k})$ *High Partial Credit:* Significant Work • Evaluates h to 1288·267 • One incorrect substitution and finished correctly • Correct answer without supporting work • Incorrect calculator mode but otherwise correct (once only) **Note:** Radian: $h = 9142$; Distance = 10392 Gradian: $h = 1281$; Distance = 2531
(b) (i)	$\frac{x}{\sin 47} = \frac{260}{\sin 36}$ $x = \frac{260(\sin 47)}{\sin 36}$ $x = 323{\cdot}5058411$ $x = 323{\cdot}506$ $x = 324$	**Scale 15D (0, 4, 7, 11, 15)** *Low Partial Credit:* Work of merit • Any work of merit to identify the remaining angle required *Mid Partial Credit:* • Sine Rule with some substitution *High Partial Credit:* Significant Work • Sine Rule fully substituted • One incorrect substitution followed by correct calculation • Correct answer without supporting work • Incorrect calculator mode but otherwise correct (once only) *Zero Credit:* • Treats triangle as right angled **Note:** Radian: $x = —32$ Gradian: $x = 327$

(b) (ii)	$$C = 2\pi(10)\left(\frac{70}{360}\right)$$ $$C = 12{\cdot}217$$ $$\text{Total Distance} = 10 + 10 + 12{\cdot}21$$ $$= 32{\cdot}217$$ $$= 32{\cdot}22$$	**Scale 10D (0, 3, 5, 8, 10)** *Low Partial Credit:* Work of merit • Reference to $\frac{70}{360}$ • Arc Length formula with some correct substitution. • Ignores C answer as $10 + 10 = 20$ *Mid Partial Credit:* • $\mathrm{C} = 2\pi(10) \times \frac{70}{360}$ *High Partial Credit:* Significant Work • Evaluates arc length to $12{\cdot}217$ • One incorrect substitution and finished correctly • Correct answer without work

Question 7 **(55 marks)**

A vertical mobile phone mast, $[DC]$, of height h m, is secured with two cables: $[AC]$ of length x m, and $[BC]$ of length y m, as shown in the diagram.
The angle of elevation to the top of the mast from A is $30°$ and from B is $45°$.



(a) **(i)** Explain why $|\angle BCA|$ is $105°$.

$30 + 45 = 75$
$180° - 75° = \underline{105°}$

(ii) The horizontal distance from A to B is 100 m.
Use the triangle ABC to find the length of y.
Give your answer correct to one decimal place.

$\frac{100}{\sin 105} = \frac{y}{\sin 30} \Rightarrow y = \frac{100(\sin 30)}{\sin 105}$

51.8 m

(iii) Using your answer to **Part (a)(ii)** or otherwise, find the value of h and the value of x. Give your answers correct to 1 decimal place.

The height of the mast, h.

$$\begin{array}{r} 180 \\ -135 \\ \hline 45^\circ \end{array}$$

$$\frac{51.8}{\sin 90} = \frac{h}{\sin 45}$$

$$h = \frac{51.8(\sin 45)}{\sin 90}$$

$$h = 36.6\text{ m}$$

The length of the cable, x.

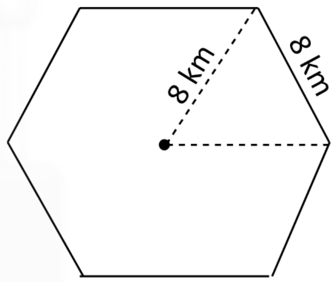
$$\frac{51.8}{\sin 30} = \frac{x}{\sin 45}$$

$$x = \frac{51.8(\sin 45)}{\sin 30}$$

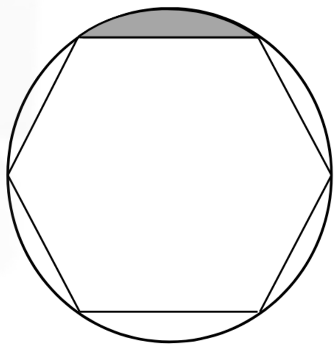
$$x = 73.3\text{ m}$$

(b) The two cables to secure the mast costs €25 per metre. The mast itself costs €580 per metre. VAT at 23% is then added in each case.
Calculate the total cost of the cables and mast after VAT is included.

(c) (i) The mast can provide a strong signal for an area in the shape of a regular hexagon of side 8 km, as shown in the diagram.
Find the area of the hexagon.
Give your answer in km^2, correct to 2 decimal places.



(ii) A circle which touches all vertices of the hexagon can show areas where the signal is weak. One of these areas is shaded in the diagram.
Find this shaded area.
Give your answer in km^2, correct to 1 decimal place.



Q7	Model Solution – 55 Marks	Marking Notes
(a) (i)	$60^{\circ} + 45^{\circ} = 105^{\circ}$ $or\ \ 180^{\circ} - 30^{\circ} - 45^{\circ} = 105^{\circ}$ $\text{or}\ \ 180^{\circ} - (30^{\circ} + 45^{\circ}) = 105^{\circ}$	**Scale 10B (0, 5, 10)** *Partial Credit:* Work of merit • Knowledge of sum of angles in a triangle = 180° shown
(a) (ii)	$\frac{\text{y}}{\sin 30} = \frac{100}{\sin 105}$ $\text{y} = \frac{100 \sin 30}{\sin 105}$ $\text{y} = 51{\cdot}7638$ $\text{y} = 51{\cdot}8\ \text{m}$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *High Partial Credit:* • Formula fully substituted
(a) (iii)	$\sin 45 = \frac{h}{51{\cdot}8}$ $\text{h} = 36{\cdot}62813$ $\text{h} = 36{\cdot}6\ \text{m}$ $\sin 30 = \frac{36{\cdot}6}{x}$ $x = \frac{36{\cdot}6}{\sin 30}$ $x = 73{\cdot}2\ \text{m}$	**Scale 10D (0, 3, 5, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *Mid Partial Credit:* • h or x found *High Partial Credit:* • Both formula fully substituted • Incorrect substitution followed by correct solution

(b)	$1{\cdot}23[25(73{\cdot}2 + 51{\cdot}8) + 580(36{\cdot}6)]$ $= 1{\cdot}23(24353)$ $= €29\,954{\cdot}19$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Work of merit • Cost of any element indicated *High Partial Credit:* • Each element excluding VAT formulated
(c) (i)	$\text{Area} = 6\left[\frac{1}{2}(8)(8)\sin 60\right]$ $= 6[27{\cdot}71281292]$ $= 166{\cdot}28\ (\text{km}^2)$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *High Partial Credit:* • Formula fully substituted • One incorrect substitution followed by correct solution
(c) (ii)	$\text{Area} = \frac{1}{6}[\pi(8^2) - 166 \cdot 28]$ $= \frac{1}{6}[34{\cdot}78192983]$ $= 5{\cdot}7969$ $= 5{\cdot}8\ (\text{km}^2)$	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* Work of merit • Formula with some substitution *High Partial Credit:* • Shaded area fully substituted